**University of Florida, Department of Mathematics**

# Analysis PhD Exam, May 2010

*Write solutions in a neat and logical fashion, giving complete reasons for all steps. Do any seven problems.*

**1.** Consider Lebesgue measure on the real line $\mathbb{R}$. Give an example for each of the following, if possible. If not possible, give a brief explanation. A sequence $f_n$ in $L^1(\mathbb{R})$ converging to an $f$ in $L^1(\mathbb{R})$... Which of these answers change if we replace $\mathbb{R}$ by the interval $[0, 1]$?

a) ...in the $L^1$ norm but not in measure,

b) ...in measure but not in the $L^1$ norm,

c) ...in the $L^1$ norm but not a.e.,

d) ...a.e. but not in measure.

**2.** Define the Hardy–Littlewood maximal function. State and prove the Hardy–Littlewood maximal theorem. (Begin by proving an appropriate covering lemma.)

**3.** This problem has two parts.

a) State Egorov's theorem.

b) Suppose $(X, \mathcal{M}, \mu)$ is a measure space with $\mu(X) < \infty$ and $f : X \to \mathbb{C}$ is measurable. Let $(f_n)$ be a sequence of integrable functions such that $f_n \to f$ a.e. Suppose further that the sequence $f_n$ is uniformly integrable, that is, for every $\epsilon > 0$ there exists a $\delta > 0$ such that if $E$ is measurable and $\mu(E) < \delta$, then

$$\int_E |f_n| \, d\mu < \epsilon.$$

Prove that $f$ is integrable and

$$\lim_{n\to\infty} \int_X |f_n - f| \, d\mu = 0.$$

**4.** Let $\mathcal{X}, \mathcal{Y}$ be Banach spaces. Suppose that $(T_n)$ is a sequence of bounded linear operators from $\mathcal{X}$ to $\mathcal{Y}$ such that $\lim_{n\to\infty} T_n x$ exists for every $x \in \mathcal{X}$. Prove that

$$Tx := \lim_{n\to\infty} T_n x$$

defines a bounded linear operator from $\mathcal{X}$ to $\mathcal{Y}$.

**5.** This problem has two parts.

a) State the Lebesgue–Radon–Nikodym theorem.

b) State and prove a version of the chain rule for Radon–Nikodym derivatives.

**6.** Let $X$ be a normed vector space. Given $x \in X$, define a linear functional $\hat{x} : X^* \to \mathbb{C}$ by $\hat{x}(f) = f(x)$. Prove that the map $x \to \hat{x}$ is an isometry from $X$ into $X^{**}$.

**7.** Let $\mathcal{H}$ be a Hilbert space and suppose $\mathcal{M}, \mathcal{N}$ are closed subspaces with $\mathcal{M} \perp \mathcal{N}$. Prove that $\mathcal{M} + \mathcal{N}$ is closed.

**8.** Suppose $(k_n)_{n=1}^{\infty}$ is a sequence of functions in $L^1(\mathbb{R})$ satisfying the following. Prove that

$$\lim_{n\to\infty} \|f - f * k_n\|_1 = 0$$

for all $f \in L^1(\mathbb{R})$. (Here $*$ denotes convolution: $(f * g)(x) := \int_{-\infty}^{\infty} f(x-t)g(t)\,dt$.)

a) $\displaystyle\int_{-\infty}^{\infty} k_n(t)\,dt = 1$ for all $n$,

b) $\displaystyle\sup_n \int_{-\infty}^{\infty} |k_n(t)|\,dt < \infty$, and

c) for each $\delta > 0$, $\displaystyle\sup_{|t|\geq\delta}\{|k_n(t)|\} \to 0$ as $n \to \infty$.